CHAPTER 9

BASIC PROBABILITY AND THEOREMS

CHAPTER ESSENCE AT A GLANCE:
Master sample space, classical and axiomatic probability definitions, Addition and Multiplication theorems, conditional probability, and Bayes' Theorem.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Random Experiments and Events:** An experiment whose outcome cannot be predicted with certainty; Sample Space S is the set of all outcomes; Events are subsets of S (Mutually exclusive: A intersection B = empty).

• **Classical vs Axiomatic Probability:** Classical definition: P(A) = m divided by n (favorable cases divided by total equally likely cases); Axiomatic: P(A) in [0, 1], P(S) = 1, P(A union B) = P(A) + P(B) for disjoint events.

• **Addition and Multiplication Theorems:** Addition: P(A union B) = P(A) + P(B) - P(A intersection B); Multiplication: P(A intersection B) = P(A) * P(B|A); for independent events, P(A intersection B) = P(A) * P(B).

• **Conditional Probability and Bayes Theorem:** Conditional probability P(A|B) updates likelihood given prior occurrence of B; Bayes' Theorem computes revised posterior probabilities based on sample evidence.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Addition Theorem} \iff P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

General Probability of Union of Two Arbitrary Events

$$\text{Conditional Probability} \iff P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad [P(B) > 0]$$

Updated Likelihood Under Conditioned Sample Space

$$\text{Independence Invariant} \iff A \perp B \iff P(A \cap B) = P(A) \times P(B)$$

Multiplicative Factorization of Independent Events

$$\text{Bayes' Theorem} \iff P(E_i \mid A) = \frac{P(E_i)\,P(A \mid E_i)}{\sum_{j=1}^{k} P(E_j)\,P(A \mid E_j)}$$

Posterior Probability Inversion Formula

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse mutually exclusive events with independent events; two mutually exclusive events with positive probability cannot be independent because if one occurs, the other cannot.

• **Alert:** In Bayes' Theorem, the sum of prior probabilities P(E1) + P(E2) + ... must equal exactly 1 because the hypotheses are mutually exclusive and exhaustive.

• **Alert:** Probability P(A) is strictly bounded: 0 <= P(A) <= 1; a negative probability or a value greater than 1 indicates a computational error.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define: (a) Mutually Exclusive Events, (b) Independent Events. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **(a) Mutually Exclusive Events:** Two events that cannot occur simultaneously in a single trial of an experiment; their intersection is null: P(A intersection B) = 0. **[1 Mark]**

• **(b) Independent Events:** Two events such that the occurrence or non-occurrence of one has zero influence on the probability of occurrence of the other: P(A intersection B) = P(A) * P(B). **[1 Mark]**

2. A card is drawn from a well-shuffled pack of 52 cards. What is the probability of getting either a King or a Spade? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Total Outcomes:** n = 52. Event A = Drawing a King; Event B = Drawing a Spade. **[0.5 Mark]**

• **Probabilities:** P(A) = 4 divided by 52; P(B) = 13 divided by 52. A intersection B = King of Spades (1 card), so P(A intersection B) = 1 divided by 52. **[1 Mark]**

• **Addition Theorem:** P(A union B) = P(A) + P(B) - P(A intersection B) = (4 divided by 52) + (13 divided by 52) - (1 divided by 52) = 16 divided by 52 = 4 divided by 13. **[1.5 Marks]**

3. A problem in statistics is given to two students A and B whose chances of solving it are 1 divided by 2 and 1 divided by 3 respectively. Find the probability that the problem will be solved. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Given Probabilities:** P(A) = 1 divided by 2, P(B) = 1 divided by 3. Events are independent. **[1 Mark]**

• **Complement Method:** Probability that A fails: P(A') = 1 - (1 divided by 2) = 1 divided by 2. Probability that B fails: P(B') = 1 - (1 divided by 3) = 2 divided by 3. **[1 Mark]**

• **Probability Problem is Solved:** P(Solved) = 1 - P(Both Fail) = 1 - [P(A') * P(B')] = 1 - [(1 divided by 2) * (2 divided by 3)] = 1 - (1 divided by 3) = 2 divided by 3. **[1 Mark]**

4. An industrial plant has two machines A and B. Machine A produces 60 percent of the output and Machine B produces 40 percent. 2 percent of items from A are defective, and 1 percent from B are defective. An item chosen at random is found to be defective. Find the probability that it was manufactured by Machine A. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Prior Probabilities:** P(E1) = 0.60 (Machine A), P(E2) = 0.40 (Machine B). **[1 Mark]**

• **Likelihoods of Defective (D):** P(D|E1) = 0.02, P(D|E2) = 0.01. **[1 Mark]**

• **Total Probability of Defective:** P(D) = P(E1)*P(D|E1) + P(E2)*P(D|E2) = 0.60 * 0.02 + 0.40 * 0.01 = 0.012 + 0.004 = 0.016. **[1 Mark]**

• **Bayes Formula:** P(E1|D) = [P(E1)*P(D|E1)] divided by P(D) = 0.012 divided by 0.016 = 12 divided by 16 = 3 divided by 4 = 0.75. **[1 Mark]**

5. State the three axioms of probability formulated by A.N. Kolmogorov. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Axiom 1 (Non-negativity):** For any event A, P(A) >= 0. **[0.5 Mark]**

• **Axiom 2 (Certainty):** For the entire sample space S, P(S) = 1. **[0.5 Mark]**

• **Axiom 3 (Additivity):** For any two mutually exclusive events A and B, P(A union B) = P(A) + P(B). **[1 Mark]**

6. If P(A) = 0.8, P(B) = 0.5, and P(B|A) = 0.4, find P(A intersection B) and P(A|B). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Intersection:** P(A intersection B) = P(A) * P(B|A) = 0.8 * 0.4 = 0.32. **[1 Mark]**

• **Conditional P(A|B):** P(A|B) = P(A intersection B) divided by P(B) = 0.32 divided by 0.5 = 0.64. **[1 Mark]**